DIMAS MAULANA FASYA

M.S. in (Architecture) – Building Energy Performance & Sustainable Tropical Design
Department of Architecture, Pusan National University, Busan, South Korea
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RESEARCH INTEREST

Building energy performance simulation | Passive and hybrid envelope systems | Natural and buoyancy-driven ventilation | Double-skin facades in tropical climates | thermal-hygric decoupling | Computational building science

EDUCATION

2024 – 2026 **Master of Science (M.S) in Architecture**
Department of Architecture, College of Engineering, Pusan National University, Busan, South Korea
Thesis: Energy Performance Optimization of Integrated Passive Envelope for Mid-Rise Office Buildings in Tropical Climate
Advisor: Prof. Seonghwan Yoon
GPA: 4.31 / 4.50

2013 – 2017 **Sarjana Arsitektur (S.Ars) (Equivalent to Bachelor of Architecture)**
Department of Architecture, Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, East Java, Indonesia
Final Project Entitled: Design of Riau Education Center for Forest Fire Disaster Mitigation with Eco-Sustainable Architecture Approach
GPA: 3.73 / 4.00 (Cum Laude)
Achievement: Best Graduate of the Department of Architecture in 2017

RESEARCH EXPERIENCE

September 2024 - present **Building Environment Science and Technology (Lab of the BEST), Department of Architecture, College of Engineering, Pusan National University –** Full Time Researcher
IRIS: 13278025
KRI: 164.125.222.174

October 2016 – January 2017 **Optimizing Roof Material as The Refrigeration and Acoustic Factor on Residential Home Building in Tropical Area - Building Science Laboratory, Department of Architecture, Maulana Malik Ibrahim State Islamic University – Research Assistant**

- Research focuses on the influence of building roof covering materials that will be used on the thermal and acoustic conditions of residential homes in tropical climates.

October 2015 – January 2016 **Optimizing Masterplan Development of Campus 3 of Maulana Malik Ibrahim State Islamic University, Malang Low Impact Development Based - Building Science Laboratory, Department of Architecture, Maulana Malik Ibrahim State Islamic University – Research Assistant**

- Research focuses on the influence of building materials (specifically building facades) that will be used on building thermal and acoustic conditions in the Development of Campus 3 Maulana Malik Ibrahim State Islamic University

WORK EXPERIENCE

June 2022 – August 2026 **Studio Humanis, Pekanbaru, Riau – Deputy Head of Studio**

- Led the exterior architectural design team from concept design through 3D visualization.

- Directed client meetings and design presentations, translating client briefs into design documentation.
- Delivered 16+ exterior and 12 interior design projects, meeting agreed technical specifications.

April 2022 – August 2023 **PT. Alva Kreatif Industri, Pekanbaru, Riau – Senior Architect**

- Managed comprehensive architectural design process for 25 high-impact projects, with a primary specialization in healthcare facilities, delivering 20 interior spaces and 5 exterior developments.
- Ensured optimal space utilization, aesthetic alignment, and strict compliance with structural requirements across all deliverables.

November 2021 – March 2022 **PT. Galepala Propertindo, Pekanbaru, Riau – Senior Architect**

- Executed architectural design and project management for 11 diverse developments, including residential houses, restaurant, and hospital facilities.
- Successfully completed 5 exterior and 6 interior projects within a highly condensed 5-month timeline without compromising design integrity.

April 2020 – January 2022 **Monthly Bulletin of Indonesian Institute of Architects of Riau Province (BULIR), Pekanbaru, Riau – Layout Editor & Distributor**

Responsible and focused on the design and layout of the newsletter section and advertisements

July 2020 – November 2021 **FCS Indonesia, Pekanbaru, Riau – Senior Architect**

- Directed the comprehensive design development for 15 projects (9 exteriors, 6 interiors), notably spearheading the long-term architectural planning for 2 large-scale government port authority complexes (KSOP).
- Navigated complex institutional requirements and governmental design standards to deliver high-profile public infrastructure and administrative offices.

February 2018 – June 2020 **Kyuu Architecture Managed by PT. Ardeko Indonesia, Pekanbaru, Riau - Architect**

- Executed architectural design for 30 diverse projects, prominently managing the multi-site corporate branch developments for United Tractors (Astra Group).
- Ensured consistent corporate design standards across 5 diverse major cities in Indonesia (Jakarta, Tarakan, Jayapura, Palembang, and Jambi).

September 2017 – February 2018 **Angkasa Architects, Pekanbaru, Riau – Junior Architect**

Assisted in the design and technical drafting phases for 8 residential and commercial interior/exterior building projects.

AWARDS

2026 **Best Presenter at the International Conference on Green Technology 2026 – Maulana Malik Ibrahim, State Islamic University, Malang, East Java – International Scale**

2018 **Top 30th Best Essay "Essay Pemuda Competition" Organized by STUDEC International – National Scale**

2016 **6th Place Best Proposal of Student Competitive Research (PKM) Organized by Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, East Java – Intra-University Scale**

2015 **3rd Place in "City Design Competition" – General Category – Organized by The Governments of Malang City, East Java – Regional Scale –** Team Name: "Splendid"

FUNDING

2023 - 2026 **Global Korean Scholarship (GKS) –** Funded by National Institute for International Education (NIIED) – Government of Republic of Korea

2024 - 2026 **BK21 FOUR –** Funded by National Research Foundation of Korea (Research and Education Program for Living SOC Innovative Design Using Empty Spaces in Declining Cities)

CONFERENCES

2026 **The 16th International Conference on Green Technology 2026 – Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, Indonesia**
Presentation/Paper Title: Quantifying the CO_2 Based Indoor Air Quality Penalty of Passive Envelope Optimization in Tropical Mid-Rise Offices

2025 **The International Conference by Indonesian Students in Korea (ICONIK) held in Kangwon National University, Chuncheon, South Korea**
Presentation / Paper Title: Optimizing Hybrid Cooling Strategies for Carbon Neutrality in Tropical Mid-Rise Office in Jakarta, Indonesia: Integrating Windcatcher and Louvers in Double Skin Façade System

2025 **International Symposium of Future Earth Collaboration of Pusan National University (PNU) G-LAMP and Laboratory of Architectural Science and Technology – Sepuluh Nopember Institute of Technology, Surabaya, Indonesia**
Presentation / Paper Title: A study on the Impact of a Hybrid Cooling System on Indoor Air Quality in a Tropical Mid-Rise Office Building in Jakarta, Indonesia

2025 **The 15th International Conference on Green Technology 2025 – Faculty of Science and Technology, Maulana Malik Ibrahim State Islamic University, Malang, Indonesia**
Presentation/Paper Title: Optimization of Hybrid Cooling System for Tropical Mid-Rise Office Building in Jakarta, Indonesia

PUBLICATION

2026 **Paper / Journal Title: Quantifying the CO_2-Based Indoor Air Quality Penalty of Passive Envelope Optimization in Tropical Mid-Rise Offices** – Atlantis Advances in Applied Sciences (Springer); Role: First Author; Status: Editorial

2026 **Paper / Journal Title: Cyberspace-Enable Ephemeral Architecture for Tropical Public Spaces: Green-Smart Resilient Infrastructure** – E3S Web of Conferences (EDP Sciences); Role: Third Author; Status: Published

2026 **Paper / Journal Title: Thermal and Hygric Limits of Integrated Passive Envelopes in Tropical Climates: A Sensible-Latent Decoupling Analysis for Mid-Rise Office Buildings** – Architectural Science Review (Taylor & Francis) (Q1 AHCI); Role: First Author; Status: Out for Review

2026 **Paper / Journal Title: Thermodynamic Limits of Integrated Passive Envelopes in High-Entalphy Tropical Climates: Decoupling Sensible Cooling Efficiency From Latent Load Constraints** – Energy & Environment (Sage Journals) (Q1 SSCI); Role: Fist Author; Status: Awaiting Reviewer Assignment

2025 **Paper / Journal Title: Assessing the Potential of Hybrid Cooling Systems for Energy Efficiency in Tropical Mid-Rise Office Building under Jakarta's Climate Condition –** IOP Conference Series: Earth and Environmental Science (Scopus Index); Role: Fist Author; Status: Published

RESEARCH PROJECT

2026 **Solar Microclimate Canopy Development (Team "Linkway")** Funded by PNU BK21 Innovation Course
Description: Developed an AI-driven, aero-kinematic building-integrated photovoltaic (BIPV_ canopy system – integrated Edge AI (ESP32) and dual-function solar sensors to automate ventury-effect passive cooling and stack effect thermal purging for urban pedestrian comfort

ARCHITECTURAL COMPETITION EXPERIENCE

2021 **Participants in Rendering Challenge 2021 organized by Andy Rahman Architects, DEMIX, and SketchUnesia – National Scale**

2019 **Participants in the Nusantara Architectural Design Competition Tourism Information Center organized by PT. Propan Raya – National Scale**
- The Design Object is the Tourism Information Center of Lake Toba, North Sumatera

2019 **Participants in the Mitra10 IDEA Architecture Design Competition themed "Contemporary Small House" organized by Mitra10 and IDEA – National Scale**
- The Design Object is a house design on a 72 m^2 design site with a maximum building area of 60 m^2

2019 **Participants in the Onduline Green Roof Award #4 with the theme "Tropical Green Roof System" organized by Onduline – National Scale**
- The Design Objects are commercial buildings consisting of a capsule hotel and eatery

2018 **Participants in the Mitra10 Architecture Design Competition themed "Tropical Contemporary House" organized by Mitra10 – National Scale**

2018 **Participants in the West Bandung Regency Gate Design Competition organized by the Government of West Bandung Regency in collaboration with the Indonesian Institute of Architects of West Java Province – National Scale**

2017 **Participants in the Architectural Design Idea of Public Park Toilet Building Competition organized by PT. American Standard Indonesia – National Scale**

PROFESSIONAL MEMBERSHIP

The Indonesian Institute of Architects (Ikatan Arsitek Indonesia) (Ordinary Member)
Registered Number: 19546 192 1200

ORGANIZATION ACTIVITIES

2023 - 2025 **The Indonesian Institute of Architects of Riau Province - Board Member – Division of Mediation and Advocacy**

2019 - 2022 **The Indonesian Institute of Architects of Riau Province - Board Member – Division of Organizational, Institutional, and International Relations**
As a division member, I am responsible for publishing organizational information to social media and public relations.

TRAINING, WORKSHOP, AND FORUM

January 25th, 2023 **Architect Professional Development 2: Architectural and Environmental Planning & Multidisciplinary Project Management** organized by the Indonesian Institute of Architects of South Sumatera Province

April 18th, 2022 **Building Structure Design Workshop for a Contoured Land** organized by the Department of Architecture, Maulana Malik Ibrahim State Islamic University
Keynote Speaker: Prof. Andrew Charleson (Victoria University of Wellington, New Zealand)

March 11th, 2022 **Building Structure Design Workshop for Disaster Resilience** organized by the Department of Architecture, Maulana Malik Ibrahim State Islamic University
Keynote Speaker: Prof. Andrew Charleson (Victoria University of Wellington, New Zealand)

July 16th, 2020 **Upgrading Strata 2 – Urban Planning** organized by the Indonesian Institute of Architects of Riau Province

July 15th, 2020 **Upgrading Strata 1 – Land Use** organized by the Indonesian Institute of Architects of Riau Province

January 26th, 2020 **Upgrading the Ethics Code and Norms of Architect Profession Attitude** organized by the Indonesian Institute of Architects of Riau Province

CERTIFICATION

February 3rd, 2026 **Certificate of Competence – Indonesian Professional Certification Authority (Construction Services Development Board)**
Competency: Assistant Architect (Expert) Lv. 7
Certificate Number: 85691 2161.01 7 00000048 2026
No. Reg.: F 2377 00048 2026 0568535 AR 01

PROFESSIONAL/ TECHNICAL SKILLS

Technical Skills:	AutoCAD, DesignBuilder + EnergyPlus, OpenStudio, Rhinoceros + Grasshopper
Design Skills:	SketchUp, V-ray Rendering for SketchUp, Enscape, Adobe Photoshop, Adobe InDesign, Adobe Illustrator, and CorelDraw
Video Editing:	Adobe Premiere
Document Preparation:	Microsoft Office, OriginPro

LANGUAGES

Bahasa Indonesia:	**Native**	
English:	**Upper-Intermediate – (TOEIC: 710) (2026)**	
	Listening Comprehension	: 400
	Reading Comprehension	: 310
Korea:	**TOPIK 2: Level 3 (2024)**	
	Listening	: 64/100
	Writing	: 27/100
	Reading	: 36/100

REFERENCES

- **Seonghwan Yoon, Ph.D (Advisor) – Master Degree**
  Department of Architecture, College of Engineering, Pusan National University
  Professor
  Tel: +82 (051) 510-2355
  Email: yoon@pusan.ac.kr
  Office: Busandaehak-ro, 63beon-gil, Geumjeong-gu, Busan, South Korea, 46241

- **Dr. Aldrin Yusuf Firmansyah, M.T. (Supervisor) – Undergraduate Degree**
  Department of Architecture, Maulana Malik Ibrahim State Islamic University, Malang, East Java – Lecturer
  Tel: +62 82143099898
  Email: aldrinfirmansyah@yahoo.com
  Office: Jalan Gajayana 50, Lowokwaru, Malang 65144

- **Ernaning Setyowati, M.T. (Laboratory Supervisor) – Undergraduate Degree**
  Department of Architecture, Maulana Malik Ibrahim State Islamic University, Malang, East Java - Head of Building Science Laboratory / Lecturer
  Tel: +62 8155552680
  Email: ernaning@arch.uin-malang.ac.id
  Office: Jalan Gajayana 50, Lowokwaru, Malang 65144